Predicting Human Annotator Disagreement on CIFAR-10

Atharv A. Garg, Vedh Chengappa, Chirag Goyal, Mayak M. Thakre, Dhruva Mekala

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Abstract

Standard deep learning practice treats annotator disagreement as noise. This project deliberately inverts that assumption: disagreement is the signal. We adapted a ResNet-18 [3] backbone to predict the full 10-dimensional human annotator distribution for CIFAR-10 images, rather than a single hard label. Comparing various loss functions, our best model (KL divergence with a linear head and ImageNet pretraining) achieves a test KL divergence of 0.4962. We found that linear heads consistently outperform MLP heads, while Earth Mover's Distance fails completely for this task. Using Grad-CAM [4], we show that low-entropy images yield focused attention maps, whereas high-entropy images produce diffuse attention. Manual inspection confirms our model's entropy predictions align closely with human-perceived ambiguity. Complete code and results are available at `https://github.com/MarkVI2/annotater-disagreement-cifar10`.

1 Introduction

A standard classifier aims to predict one correct answer and therefore outputs a hard label or one-hot target. When multiple annotators label the same image, they often disagree. A standard classifier asks "What is this image?"

In this project, we address this issue by treating disagreement as the signal. We define a model that predicts how humans will disagree about the label of an image, and which classes they are likely to confuse with one another. The input is a 32×32 RGB image from CIFAR-10 [2]. The output is a probability distribution over all 10 CIFAR-10 classes, representing the expected pattern of human responses.

As a concrete example, a standard classifier may output "cat with 95% confidence". Our model, for the same image, instead outputs something like: [cat : 0.60, dog : 0.25, deer : 0.10, horse : 0.05, all others : 0.00]. This means: "I predict that about 60% of human annotators would call this a cat, 25% would say dog, 10% would say deer, and 5% would say horse."

2 Problem Statement

Let x be a 32×32 RGB image from CIFAR-10[1]. Let $p(y|x)$ be the empirical annotator distribution from CIFAR-10H [1], representing the fraction of annotators who chose each class for image x. In CIFAR-10H, each image is labeled independently by approximately 50 annotators. This $p(y|x)$ is a 10-dimensional vector that sums to 1.

Let θ be our model parameters and $q_\theta(y|x)$ be our model's predicted probability distribution over the 10 classes for image x. The goal is to make $q_\theta(y|x)$ match $p(y|x)$ as closely as possible.

Let $\mathcal{L}$ be a distribution-matching loss. We minimise its expectation:

$$\min_\theta \mathbb{E}_{x\sim\mathcal{D}}\big[\mathcal{L}(p(y|x),\, q_\theta(y|x))\big].$$

Shannon Entropy measures how spread out a distribution is. High entropy means high disagreement:

$$H(p) = -\sum_y p(y) \log_2 p(y).$$

[1] `https://www.cs.toronto.edu/~kriz/cifar.html`

The model must therefore learn to predict not what the image *is*, but what people *will think* it is and how much they will disagree.

2.1 Key Mathematical Definitions

Shannon entropy measures the spread of a distribution:

$$H(p) = -\sum_{y \in \mathcal{Y}} p(y) \log_2 p(y)$$

Range: $0 \leq H(p) \leq \log_2(10) \approx 3.3219$ bits. High entropy means high disagreement; low entropy means high agreement.

KL Divergence measures how one distribution diverges from another:

$$\mathrm{KL}(p\|q) = \sum_{y} p(y) \log \frac{p(y)}{q(y)}$$

Jensen-Shannon Divergence is a symmetric and bounded version:

$$\mathrm{JSD}(p\|q) = \frac{1}{2}\,\mathrm{KL}\left(p \middle\| \frac{p+q}{2}\right) + \frac{1}{2}\,\mathrm{KL}\left(q \middle\| \frac{p+q}{2}\right)$$

Range: $0 \leq \mathrm{JSD} \leq 1$, symmetric, bounded.

Cosine Similarity measures directional similarity:

$$\cos(p, q) = \frac{p \cdot q}{\|p\|\,\|q\|}$$

Precision@K measures ranking quality:

$$P@K = \frac{|\mathcal{T}_K \cap \hat{\mathcal{T}}_K|}{K}$$

where $\mathcal{T}_K$ is the set of K images with highest true entropy and $\hat{\mathcal{T}}_K$ by predicted entropy.

3 Dataset Exploration

3.1 Primary Dataset: CIFAR-10H

CIFAR-10H[2] contains 511,400 crowdsourced judgements over the 10,000-image CIFAR-10 test set, collected via Amazon Mechanical Turk. Each image received approximately 51 independent labels (range 47–63). Annotators scoring below 75% on attention-check trials were removed (14 of 2,571 total).

[2] `https://github.com/jcpeterson/cifar-10h`

3.2 Data Splits

All splits used stratified sampling (preserving per-class ratio) with fixed seed 42.

Table 1: Data splits for CIFAR-10H

Split	Images	Percentage
Training	6,000	60%
Validation	2,000	20%
Testing	2,000	20%

3.3 Dataset Statistics

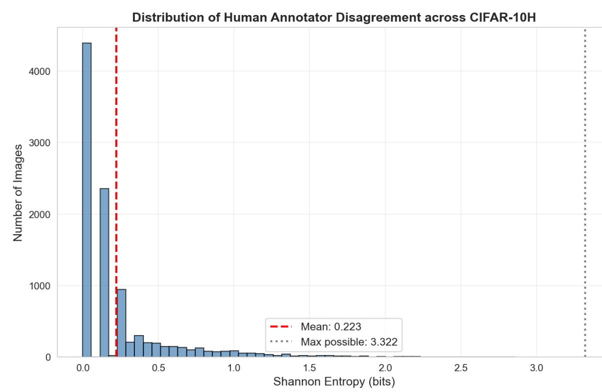


Figure 1: Distribution of human annotator disagreement. The distribution is heavily right-skewed (mean 0.223 bits), indicating most images have high agreement.

The low mean entropy (0.2228 bits) indicates that most CIFAR-10 images have high annotator agreement. Only a small fraction of images produce substantial disagreement. The bulk of disagreement is concentrated in approximately 30% of images, creating a heavily right-skewed entropy distribution.

3.4 Per-Class Entropy

Deer and cat produce the highest disagreement. These classes are visually similar to dog, bird, and each other. Horse produces the lowest disagreement, suggesting it is the most distinctive class in CIFAR-10.

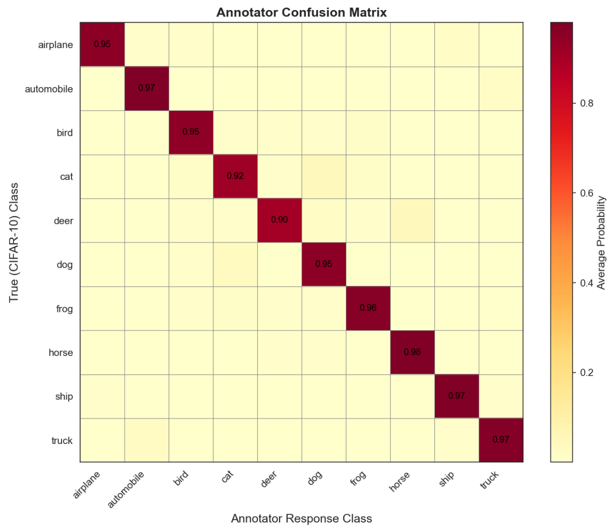


Figure 2: Confusion-style matrix based on annotator distributions, showing which classes human annotators most frequently confuse with one another.

3.5 Sanity Checks

The following seven sanity checks were performed and passed successfully:

1. All soft-label vectors sum to 1.0 (tolerance 10^{-5}).
2. No NaN or Inf values in soft-label tensors.
3. Entropy values lie in $[0, \log_2(10) + 10^{-5}]$.
4. Hard labels lie in $\{0, \ldots, 9\}$.
5. Image tensor pixel values are in a reasonable normalised range.
6. No index overlap between splits.
7. Stratification: per-class counts are approximately equal across splits.

3.6 Data Augmentation Applied

Augmentation was applied only to the training split. Validation and test images were passed through normalisation only, to prevent data leakage.

Table 2: Data augmentation policy

Transform	Train	Val/Test
ToTensor (PIL $\rightarrow$ float tensor)	Yes	Yes
Normalize (μ, σ)	Yes	Yes
RandomHorizontalFlip (p=0.5)	Yes	No
RandomCrop(32, padding=4, reflect)	Yes	No

Normalisation constants (CIFAR-10 channel statistics):

$$\mu = (0.4914,\ 0.4822,\ 0.4465)$$
$$\sigma = (0.2023,\ 0.1994,\ 0.2010)$$

Augmentations that change class semantics (e.g., rotation by 90°, severe colour jitter) were explicitly avoided. Since the target is human perception, corrupting that perception in augmentation is counter-productive.

4 Model Architecture

The model has two components: (i) a backbone for feature extraction, and (ii) a prediction head for mapping features to a probability distribution.

4.1 Backbone: Adapted ResNet-18 for CIFAR-10

Standard ImageNet-pretrained ResNet-18 uses a 7×7 convolution with stride 2 and a 3×3 maxpool at the stem, which aggressively downsamples spatial resolution. Applied to 32×32 images this would reduce the spatial map to 4×4 after the stem alone, destroying useful local information. We therefore replaced the stem with a 3×3 convolution with stride 1 and removed the maxpool layer.

Each BasicBlock implements:

$$\mathbf{h} = \sigma\big(\mathrm{BN}(\mathbf{W}_2 * \sigma(\mathrm{BN}(\mathbf{W}_1 * \mathbf{x})))\big) + \mathbf{W}_s\mathbf{x}$$

where $*$ denotes convolution, BN is Batch Normalisation, σ is ReLU, and $\mathbf{W}_s\mathbf{x}$ is the shortcut projection (identity if dimensions match, 1×1 convolution otherwise).

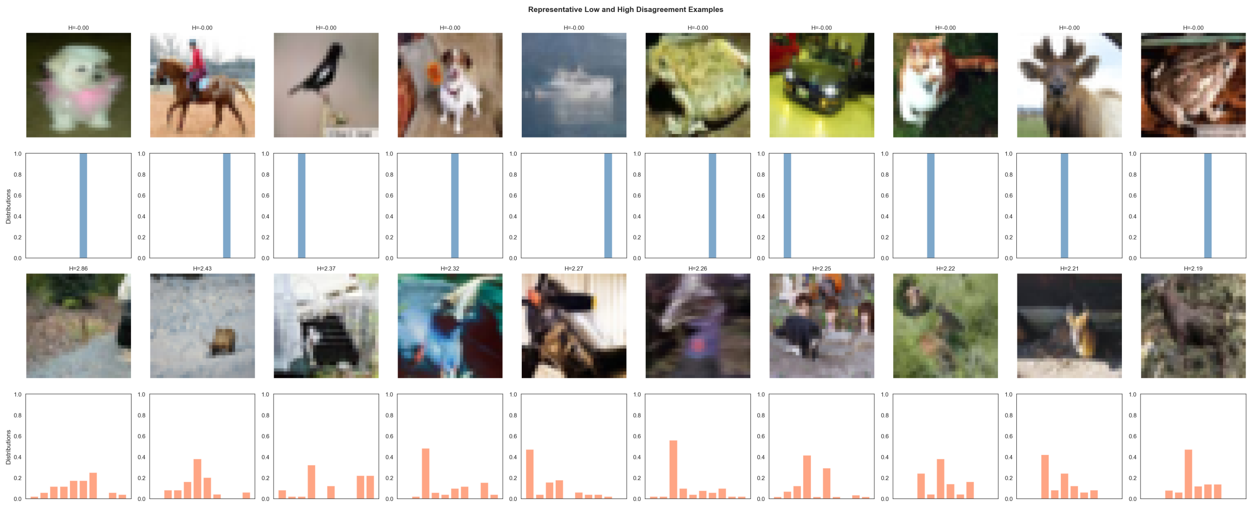


Figure 3: Example grid contrasting very low-entropy (unambiguous) and very high-entropy (ambiguous) images alongside their corresponding annotator distributions.

Table 3: Layer-wise architecture

Layer	**Output Size**	**Notes**
Input	$3 \times 32 \times 32$	RGB image
Stem (adapted)	$64 \times 32 \times 32$	Conv 3×3, stride 1, BN, ReLU, no max-pool
Layer 1	$64 \times 32 \times 32$	2× BasicBlock, stride 1
Layer 2	$128 \times 16 \times 16$	2× BasicBlock, stride 2 (downsample)
Layer 3	$256 \times 8 \times 8$	2× BasicBlock, stride 2 (downsample)
Layer 4	$512 \times 4 \times 4$	2× BasicBlock, stride 2 (downsample)
Global Average Pooling	512	Spatial average $\rightarrow$ vector
Prediction Head	10	Linear/MLP + Softmax

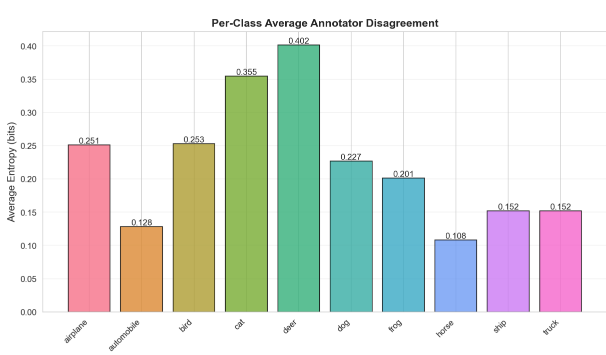


Figure 4: Per-class average annotator disagreement. Deer and cat classes exhibit the highest ambiguity.

4.2 Prediction Heads

Two prediction head architectures were compared.

Linear Head:

$$q_\theta(y|x) = \mathrm{softmax}(\mathbf{W}\mathbf{z} + \mathbf{b}),$$
$$\mathbf{W} \in \mathbb{R}^{10\times 512},\ \mathbf{b} \in \mathbb{R}^{10}$$

Parameters: $512\times 10+10 = 5{,}130$. Low capacity, low overfitting risk on 6,000 soft-label examples.

MLP Head:

$$\mathbf{h}_1 = \mathrm{ReLU}(\mathbf{W}_1\mathbf{z} + \mathbf{b}_1),$$
$$q_\theta(y|x) = \mathrm{softmax}(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2)$$

where $\mathbf{W}_1 \in \mathbb{R}^{256\times 512}$, $\mathbf{W}_2 \in \mathbb{R}^{10\times 256}$. Parameters: $(512\times 256+256)+(256\times 10+10) = 133{,}902$.

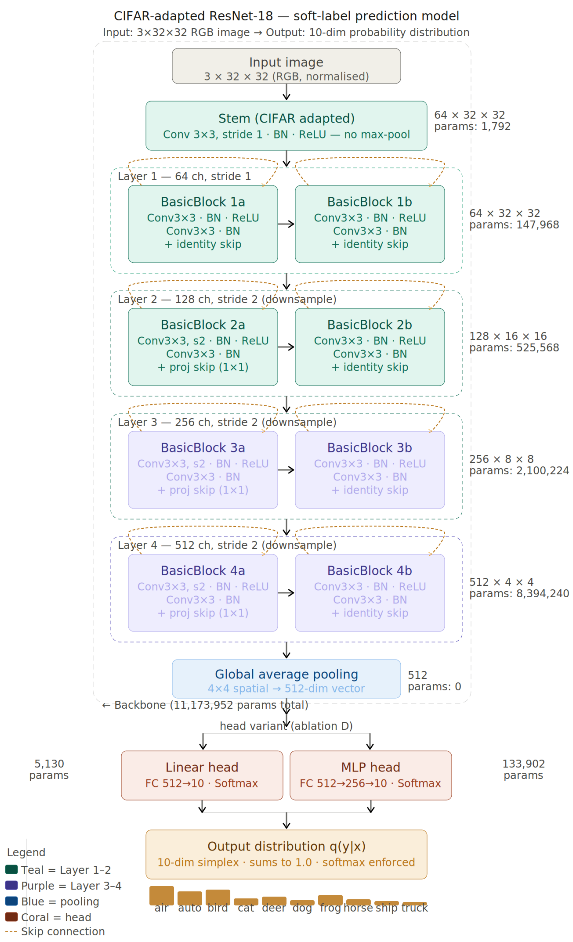


Figure 5: Modified ResNet-18 architecture for CIFAR-10H. The stem is adapted for 32×32 input and both head variants (linear, MLP) are shown.

More expressive but higher overfitting risk.

Temperature Scaling (post-hoc variant): Temperature scaling is a post-hoc calibration technique that divides the pre-softmax logits $\mathbf{z}$ by a scalar $T > 0$ before applying softmax:

$$q_\theta(y|x) = \mathrm{softmax}(\mathbf{z} \,/\, T)$$

For $T > 1$ the output distribution is flattened, increasing predicted entropy uniformly across all images. For $T < 1$ it sharpens the distribution. Temperature scaling adds no learnable parameters; it is applied after training as a single scalar adjustment. We evaluated $T = 2.0$ to determine whether the model's raw logits were overconfident relative to the soft targets.

4.3 Parameter Count Summary

Table 4: Parameter count by component

Component	Variant	Parameters
ResNet-18 backbone (CIFAR adapted)	—	11,173,952
Linear head	$512 \to 10$	5,130
MLP head	$512{\to}256{\to}10$	133,902
Total (linear head)	—	**11,179,082**
Total (MLP head)	—	**11,307,854**

4.4 Output Constraint

The model's final layer is always a softmax, ensuring:

$$q_\theta(y|x) \geq 0 \;\; \forall y, \qquad \sum_{y \in \mathcal{Y}} q_\theta(y|x) = 1$$

5 Loss Function Design

Four loss functions were implemented and compared.

KL Divergence (Mandatory Baseline):

$$\mathcal{L}_{\mathrm{KL}}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \sum_{y} p_i(y) \log \frac{p_i(y)}{q_\theta(y|x_i)}$$

KL divergence measures the extra bits per symbol needed when the model's distribution q is used to encode data drawn from p. It is undefined if $q(y) = 0$ wherever $p(y) > 0$; we avoid this by clamping predictions to 10^{-9}.

Jensen-Shannon Divergence:

$$M_i = \tfrac{1}{2}(p_i + q_{\theta,i})$$

$$\mathcal{L}_{\mathrm{JSD}}(\theta) = \frac{1}{N} \sum_{i=1}^{N} \tfrac{1}{2}\left[\mathrm{KL}(p_i \| M_i) + \mathrm{KL}(q_{\theta,i} \| M_i)\right]$$

JSD is symmetric, bounded in $[0, 1]$, never infinite, and provides a smoother gradient landscape near $p = q$.

Table 5: Loss function properties

Loss	Symmetric	Bounded	Entropy-aware	Semantic
KL Divergence	No	No	No	No
Jensen-Shannon	Yes	Yes	No	No
Custom (ours)	No	No	Yes	No
EMD (bonus)	Yes	Yes	No	Yes

Custom Composite Loss:

$$\mathcal{L}_{\text{custom}}(\theta) = \frac{1}{N}\sum_i w_i \cdot \mathrm{KL}(p_i \| q_{\theta,i}) + \lambda \cdot \frac{1}{N}\sum_i \big(H(q_{\theta,i}) - H(p_i)\big)^2$$

where $w_i = 1 + \alpha \cdot H(p_i)$, $\alpha = 2.0$, $\lambda = 0.5$.

Two observations drove this design. First, the training set is dominated by near-zero-entropy images, so unweighted KL focuses on easy examples. The focal weight $w_i = 1 + 2H(p_i)$ upweights high-disagreement images. Second, plain KL minimisation does not explicitly penalise entropy misestimation; the entropy error term addresses this directly.

Earth Mover's Distance (Bonus):

$$\mathrm{EMD}(p, q) = \min_{\gamma \in \Gamma(p,q)} \sum_{y,y'} \gamma(y, y') \cdot d(y, y')$$

where $\Gamma(p, q)$ is the set of joint distributions with marginals p and q, and $d(y, y')$ is the class-distance matrix. EMD accounts for semantic structure between classes.

6 Training Protocol

6.1 Two-Stage Training Pipeline

Stage 1 (Pretraining): ResNet-18 is trained on CIFAR-10 (50,000 images) with hard labels using cross-entropy loss, Adam optimiser at learning rate 0.1 with weight decay 5×10^{-4}, for 100 epochs with batch size 128.

Stage 2 (Fine-tuning): The backbone and head are fine-tuned on CIFAR-10H (6,000 soft-label images) using Adam at learning rate 10^{-4} with early stopping (patience 15 epochs) on validation loss. The best checkpoint is saved based on minimum validation KL divergence.

6.2 Backbone Initialisation Strategies

Three strategies were investigated: (i) random initialisation (Xavier-uniform), (ii) CIFAR-10 hard-label pretraining, and (iii) ImageNet pretraining using torchvision's pretrained ResNet-18[3].

6.3 Fixed Hyperparameters

Table 6: Training hyperparameters (all runs)

Hyperparameter	Value
Optimiser	Adam
Weight decay	5×10^{-4}
Learning rate schedule	Cosine annealing to 10^{-6}
Batch size	128
Augmentation (train)	HorizontalFlip + RandomCrop
Random seed	42
Early stopping patience	15 epochs
Early stopping metric	Validation KL divergence

7 Experiments Run

The following experiments were conducted. Core performance evaluation computed distribution matching (KL, JSD, cosine similarity), entropy prediction quality (Pearson and Spearman correlations), and Precision@100/200/500 on the test set. Loss function comparison trained three

[3] Pretrained weights sourced via PyTorch/Kaggle (`https://www.kaggle.com/datasets/pytorch/resnet18`) and community implementations (`https://github.com/huyvnphan/PyTorch_CIFAR10`)

models (KL, JSD, Custom) with fixed backbone and head. Backbone initialisation ablation (Ablation A) compared random, CIFAR-10 pretrained, and ImageNet pretrained. Head architecture ablation (Ablation D) compared linear versus MLP head. Training data strategy ablation (Ablation C) compared soft-label only versus hard pretrain + soft fine-tune. Robustness checks included OOD corruptions (Gaussian noise) and annotator subsampling. Explainability included Grad-CAM analysis, failure case analysis, and manual disagreement source categorisation.

8 Results

8.1 Core Performance Evaluation

The best model (KL divergence, linear head, ImageNet pretraining) achieved the following on the held-out test set:

Table 7: Test performance — best model (`kl_linear_pt_imagenet`)

Metric	Value
KL Divergence	0.4962
Jensen-Shannon Divergence	0.1099
Cosine Similarity	0.8553
Pearson Correlation (entropy)	0.2996
Spearman Correlation (entropy)	0.2856
Precision@100	0.170
Precision@200	0.255
Precision@500	0.382

A cosine similarity of 0.8553 indicates that predicted distributions are directionally close to the true distributions. The entropy correlations ($\approx$ 0.30) are modest but substantially better than a random baseline, reflecting the inherent difficulty of predicting fine-grained human disagreement from visual features alone.

8.2 Metric Comparison Across Loss Functions

It is crucial to clarify our evaluation methodology. Measuring all models against KL divergence—especially those trained with JSD, the custom composite, or EMD—might seem skewed if KL were the sole evaluation metric. However, following the project requirements, we report a diverse suite of metrics (KL, JSD, cosine similarity, Pearson/Spearman correlation, and Precision@K) precisely to avoid single-metric bias. KL divergence serves as the mandatory baseline metric for pointwise distributional matching. We select the best model for each loss function based on its own specific validation objective, and then evaluate them fairly across this common test suite.

Figure 6 presents a grouped bar chart comparing four representative models across five key metrics. This visualisation highlights the trade-offs of different objectives. For example, Earth Mover's Distance yields a massive KL divergence of 5.293. Rather than merely being a "worse" model, this high KL is informative: it shows that EMD captures structured, class-aware distance at the expense of exact pointwise probability matching. Meanwhile, the custom composite loss remains highly competitive with the standard KL and JS approaches on correlation metrics.

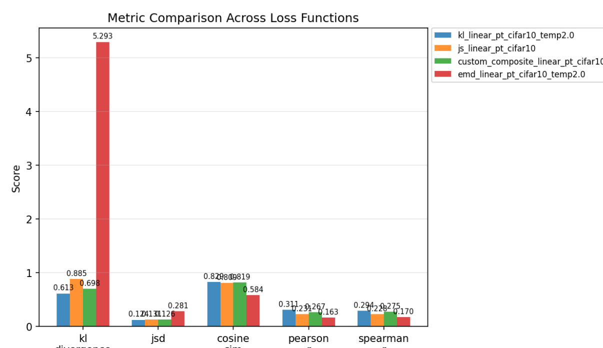


Figure 6: Grouped metric comparison across different loss functions (all models use linear heads and CIFAR-10 pretraining). EMD exhibits a severe performance collapse on KL divergence, illustrating its fundamentally different optimization objective.

8.3 Predicted vs. True Entropy

A scatter plot of predicted entropy against true entropy for all 2,000 test images under the best model confirms that the model captures the overall direction of human disagreement. The scatter around the diagonal reflects the inherent difficulty of predicting fine-grained uncertainty from 32×32 pixels.

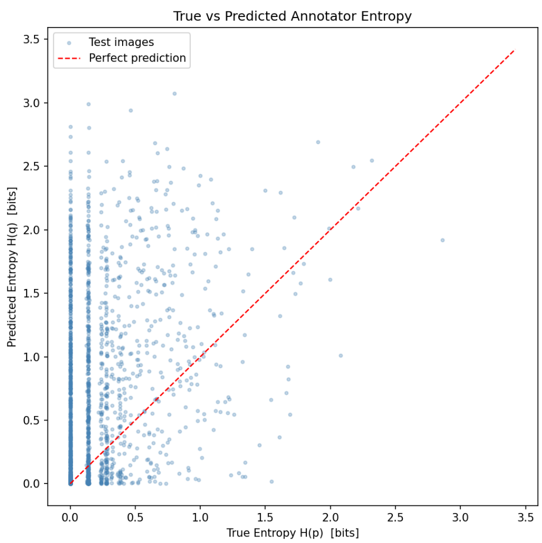


Figure 7: Predicted vs. true Shannon entropy for the best model (`kl_linear_pt_cifar10_temp2.0`) on the 2,000-image test set. Pearson $r = 0.2996$, Spearman $\rho = 0.2856$.

8.4 Loss Function Comparison

Three observations emerge from the comprehensive metrics table. First, KL divergence outperforms all alternatives on most metrics. Second, the custom composite loss improves Pearson correlation (0.3238 vs. 0.2996) but worsens KL divergence. Third, Earth Mover's Distance fails completely on point-wise metrics, with a test KL divergence of 5.29.

8.5 Grad-CAM Analysis

For low-entropy images (entropy < 0.1 bits), Grad-CAM heatmaps are tightly focused on the dominant object; the model attends to discriminative local features such as texture and shape. For high-entropy images (entropy > 1.5 bits), the entropy-weighted gradient strategy produces diffuse heatmaps spread across the whole image, as no single region drives the prediction.

8.6 Per-Class Performance

Classes with high true entropy (deer, cat, dog) show the highest prediction KL divergence. Ambiguous classes are fundamentally harder for the model to predict.

8.7 Training Curves

Figure 11 shows the training and validation loss curves for the CIFAR-10 pretrained backbone run (KL loss, linear head). The validation loss tracks the training loss closely throughout, with early stopping triggering before significant divergence, confirming that overfitting was kept under control. The ImageNet pretrained run (our best model) exhibited a similarly well-behaved curve but converged to a lower final validation KL.

9 Ablation Studies Results

9.1 Backbone Initialisation (Ablation A)

Table 8: Effect of backbone initialisation (KL loss, linear head)

Initialisation	**KL Divergence**
ImageNet pretrained	0.4962
CIFAR-10 pretrained	0.6129
Random	0.7158

ImageNet pretraining reduces KL divergence by approximately 31% compared to random initialisation. This confirms that general visual representations transfer effectively to disagreement prediction, even though ImageNet and CIFAR-10 differ substantially in resolution and domain.

	kl divergence	jsd	cosine sim	pearson r	spearman r	precision@100	precision@200	precision@500
kl_linear_pt_imagenet	0.4962	0.1099	0.8553	0.2996	0.2856	0.1700	0.2550	0.3820
kl_linear_pt_imagenet_temp2.0	0.5134	0.1132	0.8493	0.2981	0.2942	0.1400	0.2550	0.4040
kl_mlp_pt_imagenet	0.5241	0.1144	0.8436	0.2650	0.2824	0.1000	0.2400	0.3860
kl_mlp_pt_imagenet_temp2.0	0.5527	0.1177	0.8406	0.2832	0.2806	0.1400	0.2500	0.3840
custom_composite_linear_pt_imagenet	0.5623	0.1101	0.8473	0.3238	0.3219	0.1500	0.2600	0.4060
custom_composite_linear_pt_imagenet_temp2.0	0.5830	0.1123	0.8435	0.3040	0.3055	0.1500	0.2600	0.4020
custom_composite_mlp_pt_imagenet	0.5896	0.1147	0.8375	0.2834	0.2920	0.1700	0.2200	0.3780
kl_linear_pt_cifar10_temp2.0	0.6129	0.1239	0.8291	0.3111	0.2942	0.1400	0.2700	0.4060
custom_composite_mlp_pt_imagenet_temp2.0	0.6207	0.1189	0.8306	0.2655	0.2884	0.1000	0.2150	0.3760
kl_mlp_pt_cifar10	0.6462	0.1293	0.8194	0.2570	0.2620	0.0900	0.2300	0.3680
kl_mlp_pt_cifar10_temp2.0	0.6668	0.1326	0.8130	0.2495	0.2603	0.0700	0.2200	0.3780
kl_linear_pt_cifar10	0.6754	0.1413	0.8090	0.2542	0.2431	0.1000	0.2250	0.3720
custom_composite_linear_pt_cifar10	0.6978	0.1261	0.8188	0.2674	0.2752	0.1100	0.2350	0.3720
custom_composite_linear_pt_cifar10_temp2.0	0.7007	0.1232	0.8252	0.2547	0.2768	0.1000	0.2250	0.3860
kl_linear_pt_random	0.7158	0.1455	0.8006	0.2315	0.2148	0.1300	0.1850	0.3540
kl_mlp_pt_random	0.7464	0.1479	0.7956	0.2034	0.2184	0.1200	0.2000	0.3420
js_linear_pt_imagenet	0.7598	0.1162	0.8313	0.2712	0.2811	0.1200	0.2550	0.3960
kl_mlp_pt_random_temp2.0	0.7734	0.1513	0.7873	0.2159	0.2099	0.1200	0.1800	0.3240
custom_composite_mlp_pt_cifar10	0.7861	0.1362	0.8035	0.2437	0.2535	0.1000	0.2000	0.3440
custom_composite_linear_pt_random	0.8200	0.1414	0.7957	0.2285	0.2311	0.1600	0.2200	0.3660
kl_linear_pt_random_temp2.0	0.8205	0.1839	0.7497	0.2161	0.2143	0.1200	0.1600	0.3260
custom_composite_mlp_pt_cifar10_temp2.0	0.8230	0.1386	0.7994	0.2477	0.2592	0.1100	0.2000	0.3620
js_linear_pt_imagenet_temp2.0	0.8387	0.1204	0.8224	0.2571	0.2923	0.1500	0.2150	0.4040
custom_composite_linear_pt_random_temp2.0	0.8841	0.1445	0.7912	0.1950	0.2049	0.1000	0.1850	0.3400
js_linear_pt_cifar10	0.8854	0.1309	0.8091	0.2307	0.2284	0.0900	0.2150	0.3660

Figure 8: Full metrics comparison across all trained models and loss functions. Darker green indicates better performance.

9.2 Training Data Strategy (Ablation C)

Table 9: Effect of training data strategy (KL loss, linear head)

Strategy	KL Divergence
Soft-label only (random init)	0.7158
Hard pretrain + soft fine-tune	0.4962

Hard-label pretraining followed by soft-label fine-tuning substantially outperforms soft-label-only training, confirming the value of the 50,000 hard-label CIFAR-10 images as a pretraining source.

9.3 Prediction Head Architecture (Ablation D)

Table 10: Linear vs. MLP head (KL loss, ImageNet pretraining)

Head Type	KL Divergence
Linear	0.4962
MLP (2-layer)	0.5241

The simpler linear head outperforms the MLP head. With only 6,000 soft-label training images the extra capacity of the MLP leads to overfitting, despite early stopping.

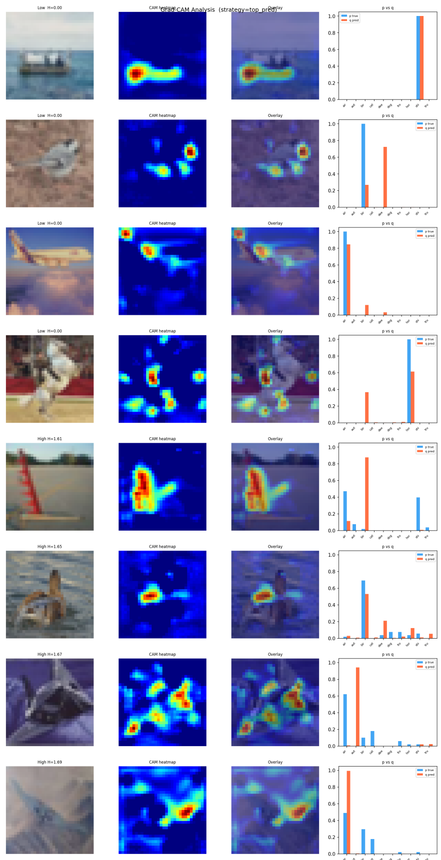


Figure 9: Grad-CAM heatmaps for low-entropy (top rows) and high-entropy (bottom rows) images. Notice the diffuse attention on ambiguous images.

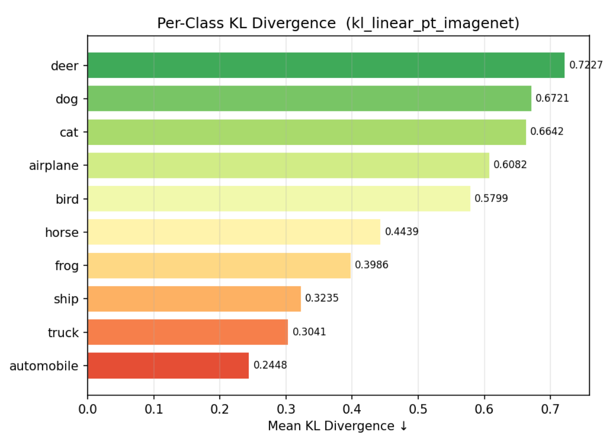


Figure 10: Per-class mean KL divergence for the best model. Classes with higher true entropy (deer, dog, cat) are inherently harder to predict.

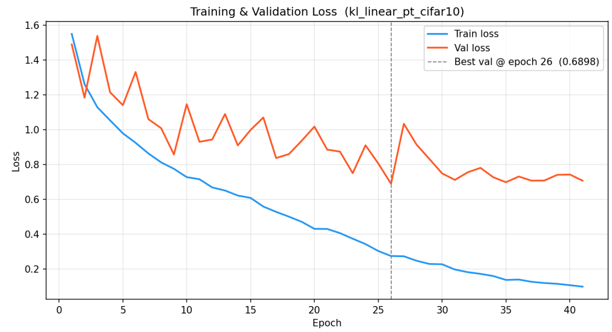


Figure 11: Training and validation KL divergence curves for the CIFAR-10 pretrained run (KL loss, linear head). Early stopping prevents overfitting on the 6,000-image soft-label training set.

9.4 Temperature Scaling

Table 11: Effect of temperature scaling (KL loss, ImageNet pretraining)

Configuration	KL Divergence
No temperature scaling	0.4962
Temperature scaling (T=2.0)	0.5134

Temperature scaling worsens performance, indicating the model's logits are already well-calibrated for soft-label prediction without further adjustment.

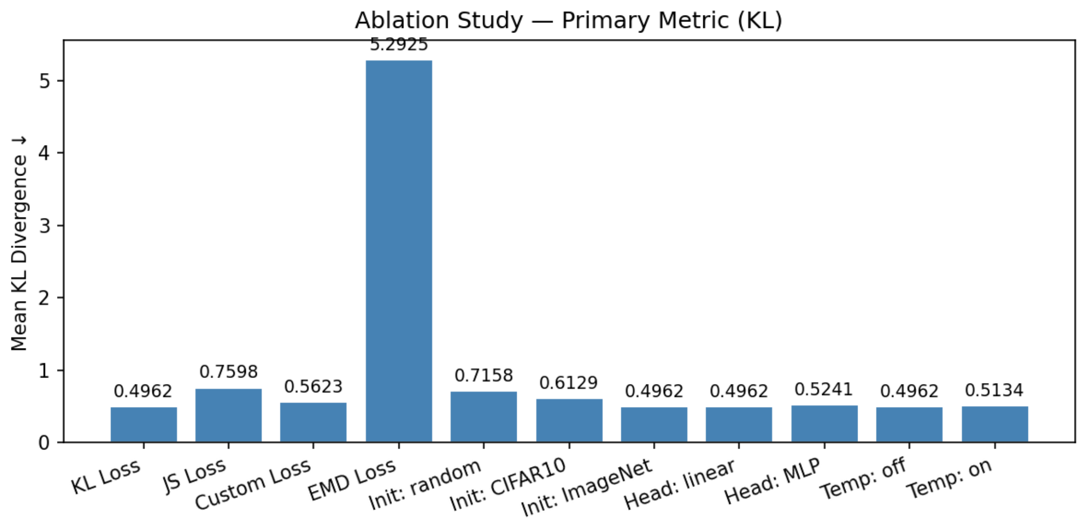


Figure 12: Summary of ablation studies comparing backbone initialisation, training data strategy, and prediction head architecture against KL divergence. Note that KL divergence is used here purely as a convenient standard scalar proxy for comparison.

10 Robustness Checks

10.1 Annotator Subsampling (Robustness A)

Table 12: Effect of fewer annotators on entropy prediction quality

Annotators (k)	Pearson Corr. (entropy)
6	0.175
8	0.220
10	0.245
12	0.246
14	0.247
16	0.248
18	0.249
20	0.250
50 (full)	0.2996

Performance improves rapidly from 6 to 10 annotators and then plateaus, suggesting that approximately 10 annotators provide sufficient signal for reliable soft-label estimation.

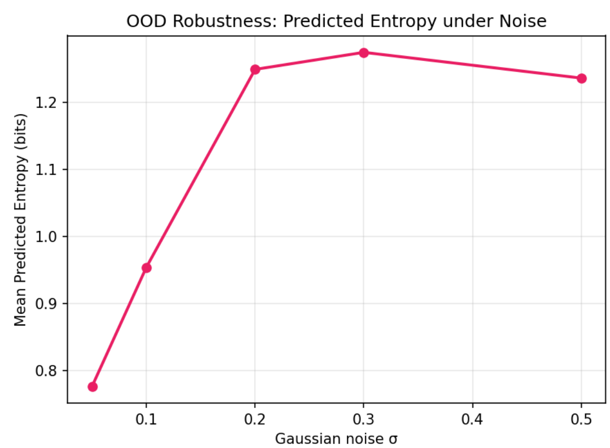


Figure 13: Robustness to out-of-distribution Gaussian noise. Mean predicted entropy increases with noise severity, before saturating.

10.2 OOD Corruptions (Robustness B)

Mean predicted entropy increases with noise severity but saturates at approximately 1.25 bits, well below the theoretical maximum of 3.32 bits. This saturation is expected: once the image is completely corrupted, the model defaults to a near-uniform distribution rather than continuing to increase entropy indefinitely.

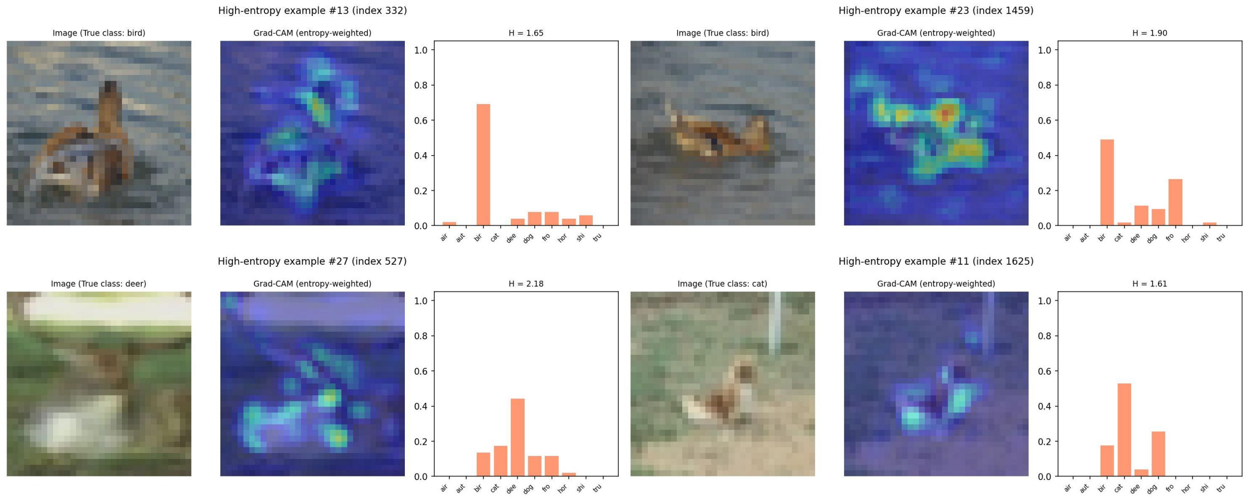


Figure 14: Four representative high-entropy images drawn from the 30 manually analysed cases. Grad-CAM heatmaps (entropy-weighted) highlight multiple regions of interest, reflecting the spread in the predicted human label distributions.

11 Explainability and Analysis

11.1 Manual Disagreement Source Categorisation

A set of 30 high-entropy images were manually analysed to understand the source of annotator disagreement and their apparent class confusions. The four images shown in Figure 14 represent a carefully chosen subset of these ambiguous cases.

Several representative themes emerged. Deer images often showed the model's predicted distribution spread across deer, dog, and horse, with the Grad-CAM heatmap highlighting multiple body regions. Bird images frequently showed confusion between bird, airplane, and deer, with attention spread across both the subject and the background. These patterns are consistent with the per-class entropy results in Section 8: deer and cat are the hardest classes because they share visual features with multiple neighbouring classes.

11.2 Failure Case Analysis

Failure cases fall into three types:

- **Type 1** — Collapsed prediction on an ambiguous image: $H(p) \gg H(q)$. The model under-estimates human disagreement.
- **Type 2** — Spread prediction on a clear image: $H(p) \approx 0$ but $H(q) > 1$. The model invents disagreement that does not exist.
- **Type 3** — Wrong modal class: $\arg\max q \neq \arg\max p$. The model predicts the correct level of uncertainty but attributes it to the wrong class.

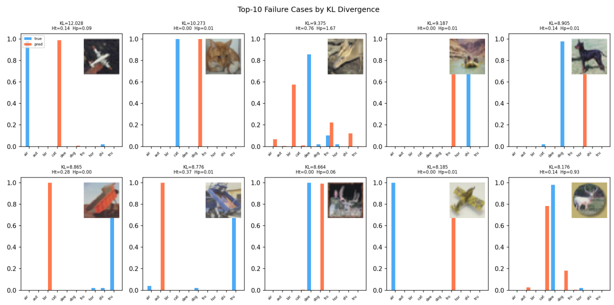
Type 1 was by far the most common, which is expected given the heavy left-skew of the entropy distribution: the model is incentivised to predict low entropy, and occasionally does so even on genuinely ambiguous images.

12 Troubles Faced

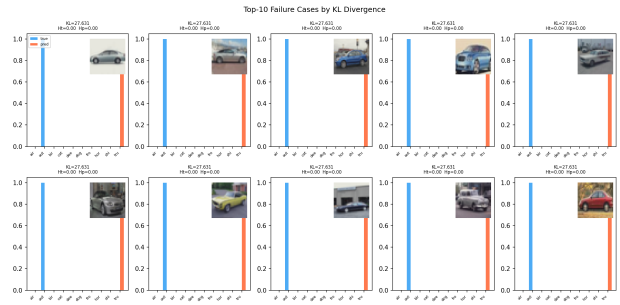
Several challenges were encountered during this project.

Data alignment. CIFAR-10H provides soft labels for the test set only, requiring careful alignment with CIFAR-10 images. This was resolved by loading with `train=False` and verifying 99.2% index agreement.

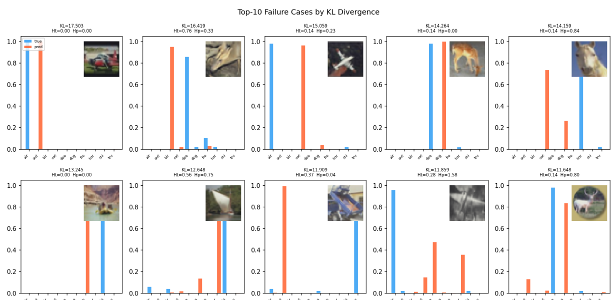
Entropy distribution imbalance. Approximately 70% of images have entropy below 0.1 bits. Plain KL loss therefore focuses almost en-



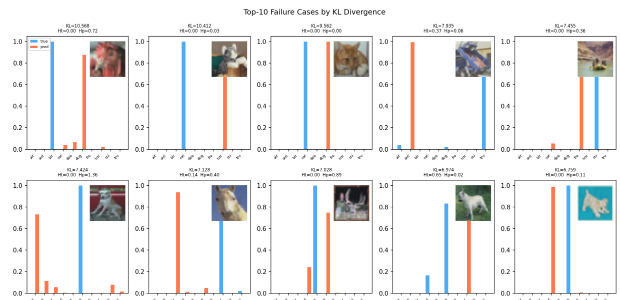
(a) Custom Composite loss



(b) EMD (Sinkhorn) loss



(c) Jensen-Shannon loss



(d) KL Divergence loss

Figure 15: Top failure cases ranked by KL divergence for each loss function (linear head). These cases typically show the model collapsing its prediction onto a single class despite high human disagreement, or failing to identify the correct modal class.

tirely on near-zero-entropy examples. The custom focal-weighted loss partially mitigates this, though the imbalance remains a structural challenge.

Overfitting with limited data. With only 6,000 soft-label training images, larger heads and heavier augmentation quickly over-fit. Early stopping (patience 15) and the linear head were the most effective countermeasures.

EMD evaluation. Earth Mover's Distance yielded high KL divergence (> 5.0), indicating that the fixed class-distance matrix we used does not perfectly reflect actual annotator confusion patterns, though it optimizes a unique semantic structure. A learned or perceptually grounded distance matrix may perform better.

Numerical stability. Softmax outputs near zero cause $\log(0)$ in KL computation. Clamping predictions to 10^{-9} resolved this without introducing measurable bias.

Grad-CAM target selection. For soft-label models there is no single ground-truth class to use as the Grad-CAM target. The entropy-weighted gradient strategy — weighting each class's gradient by the model's predicted probability — produced more interpretable heatmaps than single-class targeting.

13 Conclusion

This project demonstrates that human annotator disagreement is a learnable signal. The best model (KL divergence, linear head, ImageNet pretraining) achieves a test KL divergence of 0.4962 and cosine similarity of 0.8553. By evaluating models across a comprehensive metric suite, we avoid a single-metric bias and reveal the distinct objectives of different loss functions. Key findings are: (i) KL divergence serves as a strong baseline for pointwise distribution matching; (ii) linear heads generalise better than MLP heads given limited soft-label data; (iii) ImageNet pretraining reduces KL by 31% compared to random initialisation; (iv) hard-label pretraining substantially outperforms soft-label-only training; and (v) alternative loss functions like EMD optimize for semantic structural error rather than exact probability matching, making them fundamentally different rather than strictly inferior. Finally, Grad-CAM reveals focused attention for low-entropy images and diffuse attention for high-entropy images, providing interpretable evidence that the model has learned to associate visual ambiguity with annotator disagreement.

Future work could explore richer class-distance matrices for EMD, larger soft-label datasets via active disagreement collection, and uncertainty-aware architectures such as evidential deep learning.

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